

Layer 1 — Mobile app (React.js + Capacitor + PWA · what the engineer installs and uses)

React 18 + Vite

Component-based UI
All 9 screens built here
Fast HMR dev server

Tailwind CSS

Mobile-first utility classes
Native app look and feel
No separate CSS files

Capacitor

Wraps React build → APK/IPA
Camera + file picker access
npx cap sync → build APK

PWA Manifest

Install from Chrome browser
Service Worker caching
Works offline (app shell)

Install: [react](#) [react-dom](#) [vite](#) [tailwindcss](#) [postcss](#) [autoprefixer](#) [@capacitor/core](#) [@capacitor/android](#) [@capacitor/cli](#) [vite-plugin-pwa](#) [react-router-dom](#)

Layer 2 — State management, forms, offline SQLite (the intelligence of the app)

Zustand

Global state store
Auth · indent · sync state
Persists across screens

React Hook Form + Zod

All indent forms
Field-level validation
Server error mapping

Capacitor SQLite

On-device database
Stores all LN master data
Offline indent queue

Axios

REST API calls
JWT Bearer interceptor
Offline queue interceptor

Install: [zustand](#) [react-hook-form](#) [zod](#) [@capacitor-community/sqlite](#) [axios](#)

Layer 3 — Backend API (.NET 8 · ASP.NET Core Web API · the server your team builds)

ASP.NET Core

Web API controllers
REST endpoints (CRUD)
CORS + middleware

Entity Framework Core

ORM + code-first models
DB migrations (EF CLI)
LINQ queries

JWT Bearer Auth

Secure API tokens
PIN → token on login
5-min expiry enforced

Swagger / OpenAPI

Auto API documentation
Test endpoints in browser
Team reference

.NET pkgs: [Microsoft.AspNetCore.Authentication.JwtBearer](#) [Microsoft.EntityFrameworkCore.SqlServer](#) [Swashbuckle.AspNetCore](#) [Azure.Storage.Blobs](#) [StackExchange.Redis](#)

Layer 4 — Data stores (server-side persistence, caching, file storage)

MS SQL Server	Redis	Azure Blob Storage
Primary server database Indents · engineers · audit log Integrates with Infor LN natively	Cache LN master data (24h TTL) Avoid hitting LN on every request StackExchange.Redis NuGet pkg	Store item attachments (photos/PDF) Max 5 MB per file enforced Azure.Storage.Blobs NuGet pkg

Layer 5 — Infor LN ERP integration (12 OData sessions fetched by .NET backend)

LN OData — master data (read)	LN OData — indent write-back
tppdm6100m000 Projects · tppss2100m000 Activities tppdm6136m000 Locations · tdipu0181m000 Items whwmd2500m000 Warehouses · tssoc2100m100 Service orders whwmd3500m000 WH Locations · tssoc2600m300 Rental orders tppdm6149m000 Responsibility · tccom4134m000 Delivery pts	POST indent to LN on submission Returns In_indent_id reference Delta sync reads approval status Bi-directional via OData REST

Layer 6 — DevOps and infrastructure (Azure hosted · automated CI/CD · monitoring)

Azure App Service	Docker	GitHub Actions	Azure App Insights
Host .NET 8 API Auto-scale on load Staging + production slots	Containerise .NET 8 API Consistent dev → prod docker-compose for local dev	CI/CD pipeline Build + test + deploy On push to main branch	Monitor API errors Sync failure alerts Performance dashboards

Complete package reference — all layers

React app	react · react-dom · react-router-dom · vite
Styling	tailwindcss · postcss · autoprefixer
Native mobile	@capacitor/core · @capacitor/android · @capacitor/cli
PWA / offline	vite-plugin-pwa
State	zustand
Forms	react-hook-form · zod
Local DB	@capacitor-community/sqlite